

JACKSON C. TURNER

RESEARCH SCIENTIST | WAVES & NONLINEAR PHENOMENA

Ph.D., Applied Mathematics, Columbia University

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Experience

Harmoniqs

Brooklyn, NY

Quantum Research Scientist

Jun 2026 – Present

- Using gauge symmetries, dimension-reduction, fast algorithms, and Fourier analysis to figure out the best way to move atoms around without heating them
- Leading technical discussions with academic and industry partners; contributing to grants and investor materials
- Building human-first AI tools that allow researchers to carry out experiments on hardware

Atlas Fusion

Remote

Mathematical Physicist (Contract)

Feb – Jun 2026

- Trained differentiable ML models on spectral data of plasma equilibria to find useful plasma shapes

Columbia University, APAM Department

New York, NY

Graduate Research Assistant

May 2022 – Jun 2026

- Proved nonlinear bound states bifurcate from scattering and transmission resonances (*Nonlinearity*)
- Derived sharp orbital stability and instability criteria; Julia simulations probe beyond the theory (arXiv:2610.01875)
- Proved global variational, stability, and optimization results on mass-constrained energy functional

Zap Energy, Theory & Modeling Division

Everett, WA

Research Intern

Jun – Aug 2025

- Built a Julia MHD eigensolver with pseudo-arclength continuation; $720\times$ faster than the prior pipeline (4 h $\rightarrow$ 20 s)
- Identified novel wave instabilities that dominate in idealized model of the fusion device (arXiv:2608.00291)

Education

Columbia University

June 2026

Ph.D. in Applied Mathematics

New York, NY

Engineering Presidential Fellow. Dissertation: Resonance-induced nonlinear bound states and their orbital stability. Advisor: M. I. Weinstein.

University of Utah

May 2021

Honors B.S. in Applied Mathematics & Physics

Salt Lake City, UT

Presidential Scholar.

Technical Skills

Languages & Tools Julia (primary), Python (NumPy, SciPy), MATLAB, Git/GitHub, LaTeX

Numerical Methods Split-operator and spectral PDE solvers, nonlinear eigenproblems, pseudo-arclength continuation

Optimization & Control Quantum optimal control, adjoint gradients, L-BFGS, augmented-Lagrangian min-time control

Theory Spectral and scattering theory, nonlinear Schrödinger/Gross–Pitaevskii, stability and bifurcation, ideal MHD

Publications

- Crews, D. W. & J. C. Turner. “On the ideal stability of the sheared-flow Z pinch.” arXiv:2608.00291, 2026.
- Turner, J. C. & M. I. Weinstein. “Stability and instability analysis of resonance-induced nonlinear bound states.” arXiv:2610.01875, 2026.
- Turner, J. C. & M. I. Weinstein. “Resonance-induced nonlinear bound states.” *Nonlinearity* 39 (2026), 095030.
- Kao, C.-Y., B. Osting & J. C. Turner. “Flat Tori with Large Laplacian Eigenvalues.” *SIAM J. Appl. Algebra Geom.*, 2023.
- Turner, J. C., E. Cherkhev & D. Wang. “Expansion Method for Laplace–Beltrami Eigenfunctions.” arXiv:2210.10982, 2022.

Additional

Honors Invited talk, Simons Collaboration on Extreme Wave Phenomena; 2025 Best Poster, Alabama–Florida DE Conf.

Leadership & Languages Led teams of 10–20 volunteers across Hong Kong; fluent in Cantonese